

Descriptive Statistics

2025-11-07

1. Data Overview

1.1 Description of the Data Gathering Process

The data considered in the project is from OpenAlex. we use a pair-based keyword search to identify articles which use or construct game-theoretical models. The keywords we are taken from two lists: 1. A list of JEL codes defining the subject of study, i.e. game theory. 2. A list of terms describing game setup which ensures that we only cover papers employing or constructing game-theortic models. For each pair of keywords, we identify articles based on the appearance of keyword pairs in the title or abstract of the paper (maybe we also want to consider full texts? This would increase the number of articles drastically.) Currently, we focus only on articles and reviews written in English (Should we include books?). We drop articles from the corpus without any reference information. As the number of articles is rather low prior to 1980, we choose this year as the starting point for our analysis.

1.2 Search Terms

Table 1: List of Query Terms

Term	Term
game theory	equilibrium
game-theoretic	equilibria
cooperative game	solution concept
cooperative games	solution concepts
noncooperative game	payoff matrix
noncooperative games	pay-off matrix
non-cooperative game	payoff matrices
non-cooperative games	pay-off matrices
stochastic game	strategy
stochastic games	strategies
dynamic game	strategic
dynamic games	preference
evolutionary game	preferences
evolutionary games	player
repeated game	players
repeated games	
bargaining theory	
matching theory	

1.3 Descriptive Summary of the Data

Table 2: Descriptive Summary

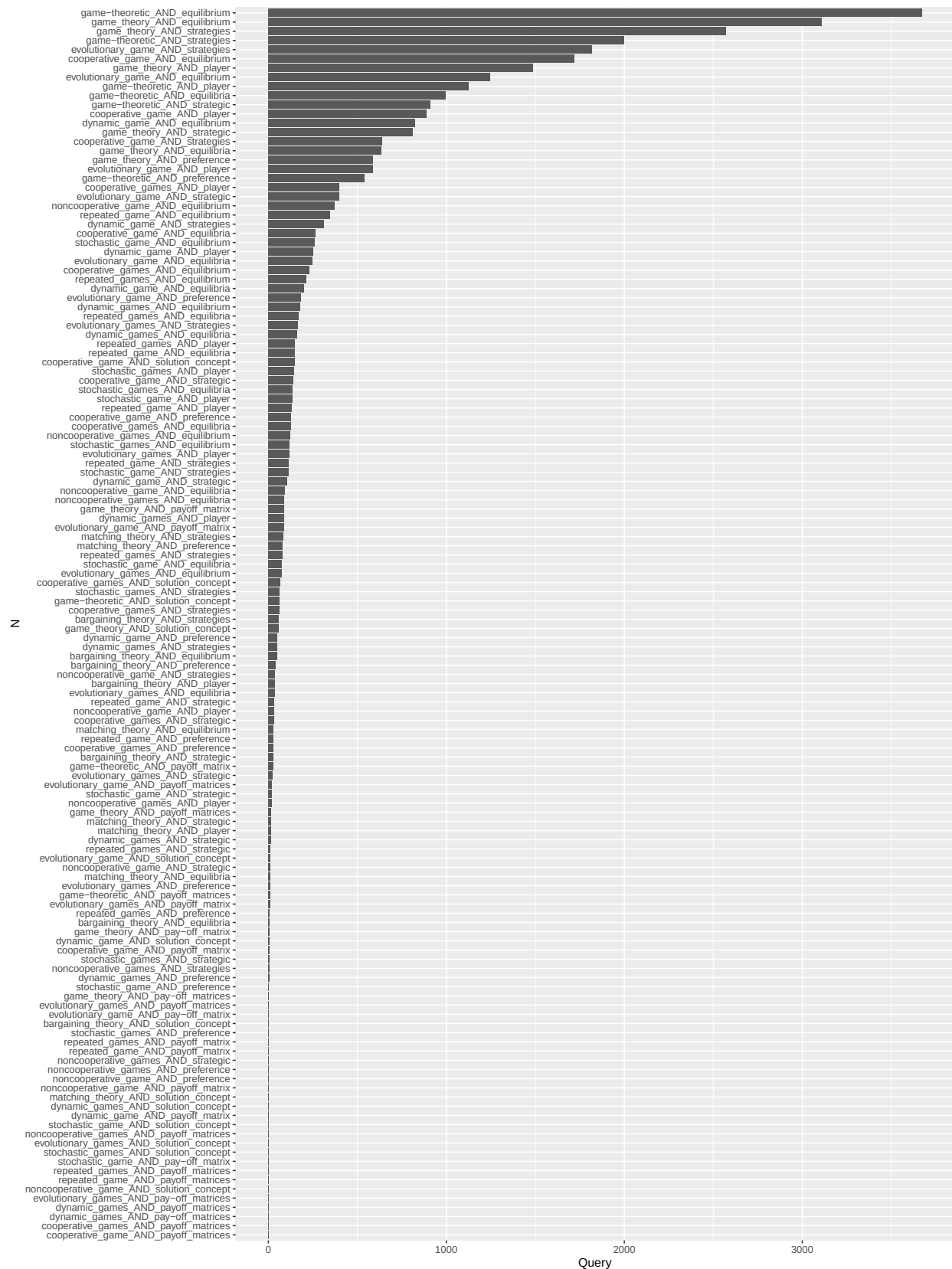
Description	
Articles	34676.00
Year (Begin)	1980.00
Year (End)	2024.00
Average Citations per Article	24.53
Authors	53383.00
Articles per Author	1.85
Authors per Article	2.85

1.4 Top Articles

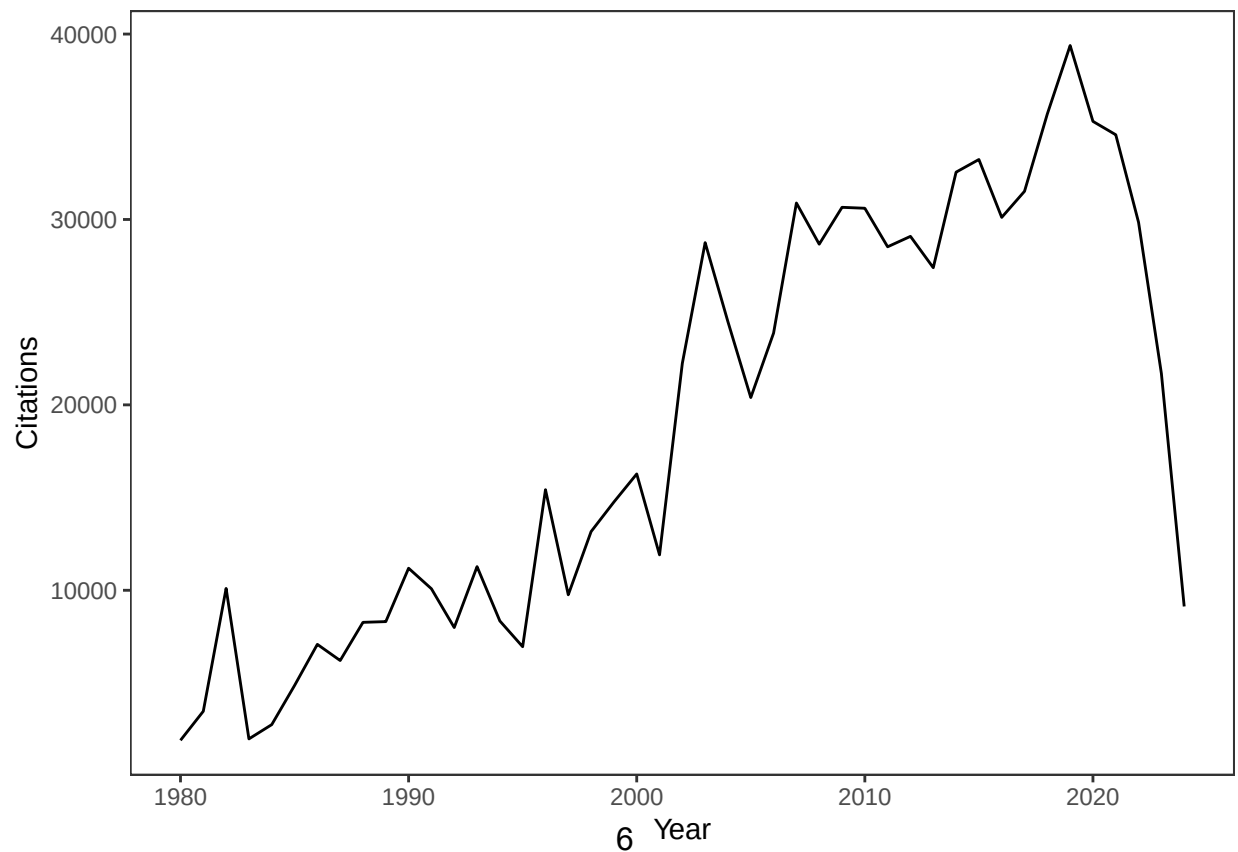
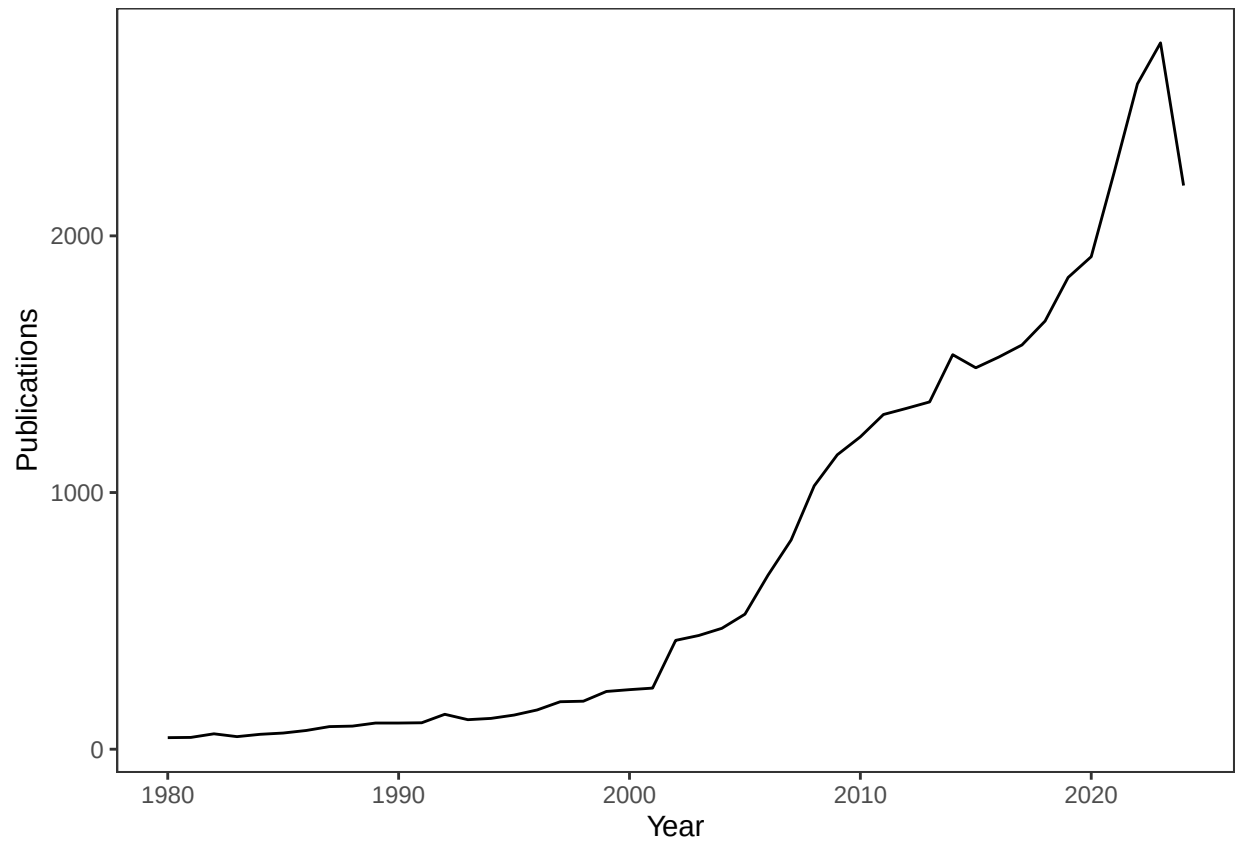
Table 3: Top Articles in Terms of Citations

Authors	Title	Year	Citations	Field
David Kempe;Jon Kleinberg;Éva Tardos	Maximizing the spread of influence through a social network	2003	7088	Physics and Astronomy
David M. Kreps;Robert Wilson	Reputation and imperfect information	1982	3101	Decision Sciences
György Szabó;Gábor Fáth	Evolutionary games on graphs	2007	2823	Social Sciences
Amir-Hamed Mohsenian-Rad;Vincent W. S. Wong;Juri Jatskevich;Robert Schober;Alberto Leon-Garcia	Autonomous Demand-Side Management Based on Game-Theoretic Energy Consumption Scheduling for the Future Smart Grid	2010	2711	Engineering
Arvind Parkhe	STRATEGIC ALLIANCE STRUCTURING: A GAME THEORETIC AND TRANSACTION COST EXAMINATION OF INTERFIRM COOPERATION.	1993	2581	Business, Management and Accounting
Xu Chen;Lei Jiao;Wenzhong Li;Xiaoming Fu	Efficient Multi-User Computation Offloading for Mobile-Edge Cloud Computing	2015	2552	Computer Science
David M. Kreps;Paul Milgrom;John Roberts;Robert Wilson	Rational cooperation in the finitely repeated prisoners' dilemma	1982	2512	Social Sciences
Drew Fudenberg;Eric Maskin	The Folk Theorem in Repeated Games with Discounting or with Incomplete Information	1986	2371	Decision Sciences
Hisashi Ohtsuki;Christoph Hauert;E. James Lieberman;Martin A. Nowak	A simple rule for the evolution of cooperation on graphs and social networks	2006	2155	Social Sciences
Charles F. Manski	Economic Analysis of Social Interactions	2000	2052	Economics, Econometrics and Finance
Martin Dufwenberg;Georg Kirchsteiger	A theory of sequential reciprocity	2003	1902	Social Sciences
Ido Erev;Alvin E. Roth	Predicting How People Play Games: Reinforcement Learning in Experimental Games with Unique, Mixed Strategy Equilibria	1998	1822	Social Sciences
NA	Rationalizability, Learning, and Equilibrium in Games with Strategic Complementarities	1990	1816	Decision Sciences
Daniel Friedman	Evolutionary Games in Economics	1991	1797	Economics, Econometrics and Finance
Barry R. Weingast	The Political Foundations of Democracy and the Rule of the Law	1997	1739	Economics, Econometrics and Finance

1.5 Number of Articles per Query

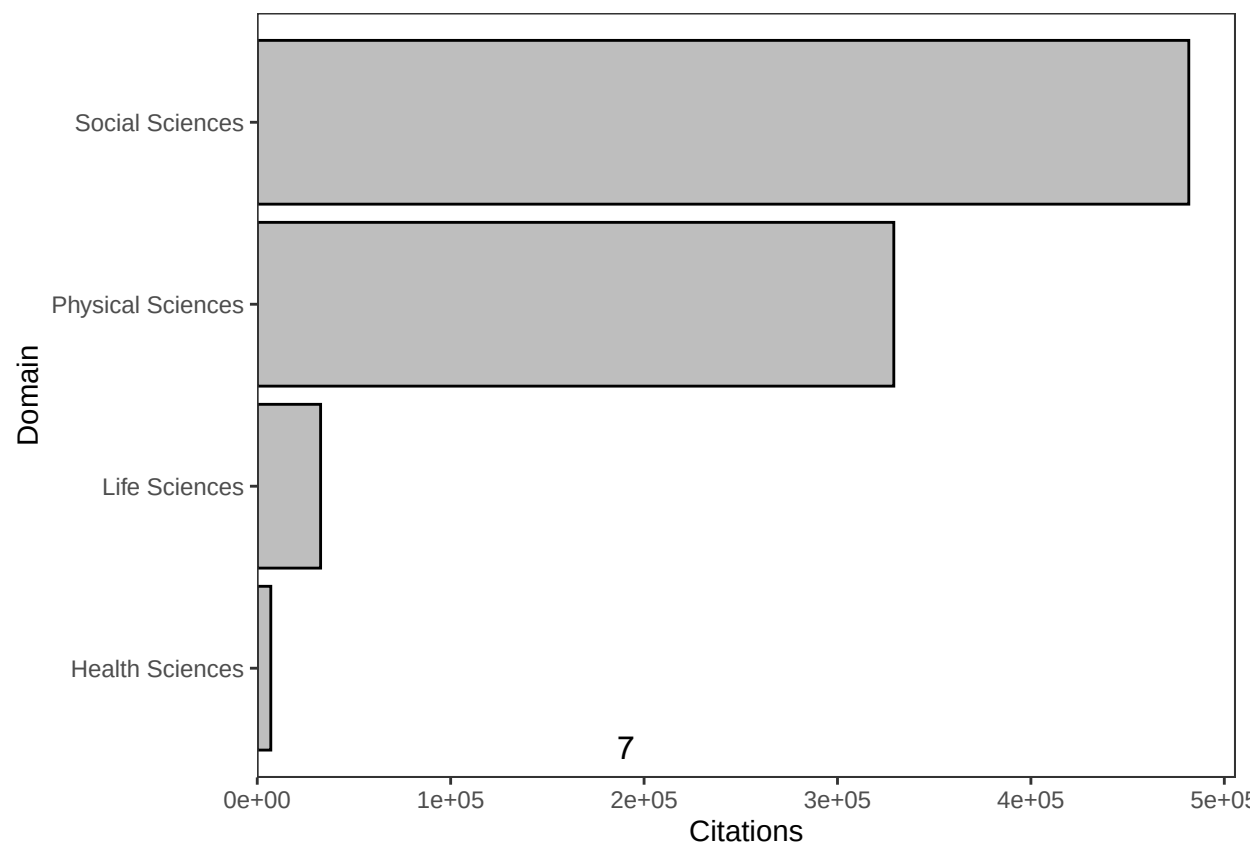
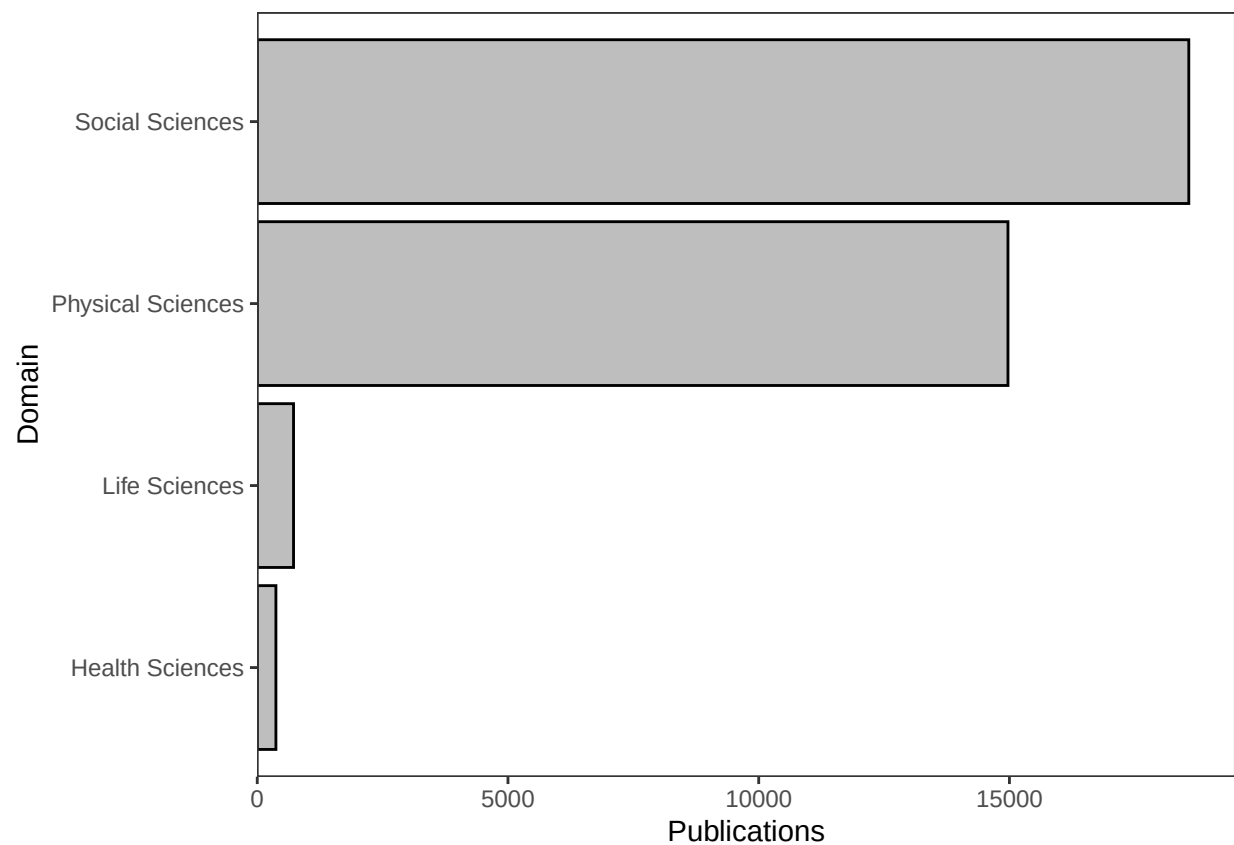


2. Publications over Time

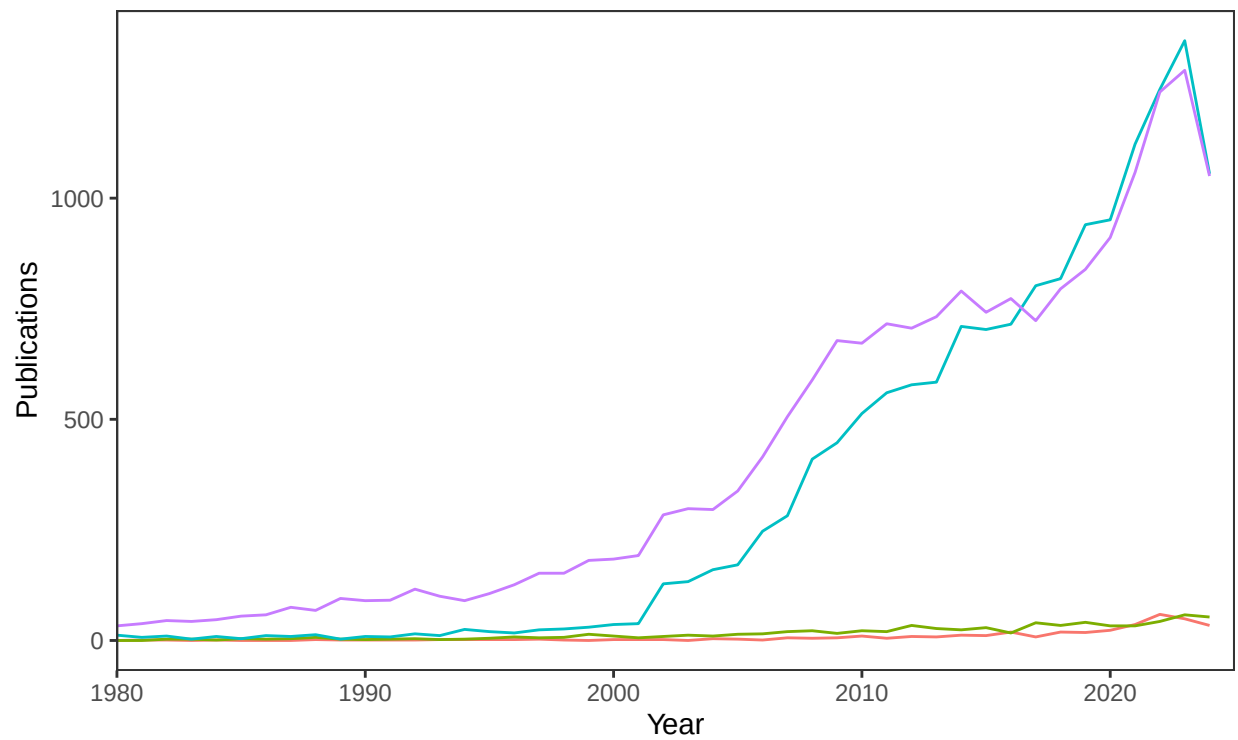


3. Domain Overview

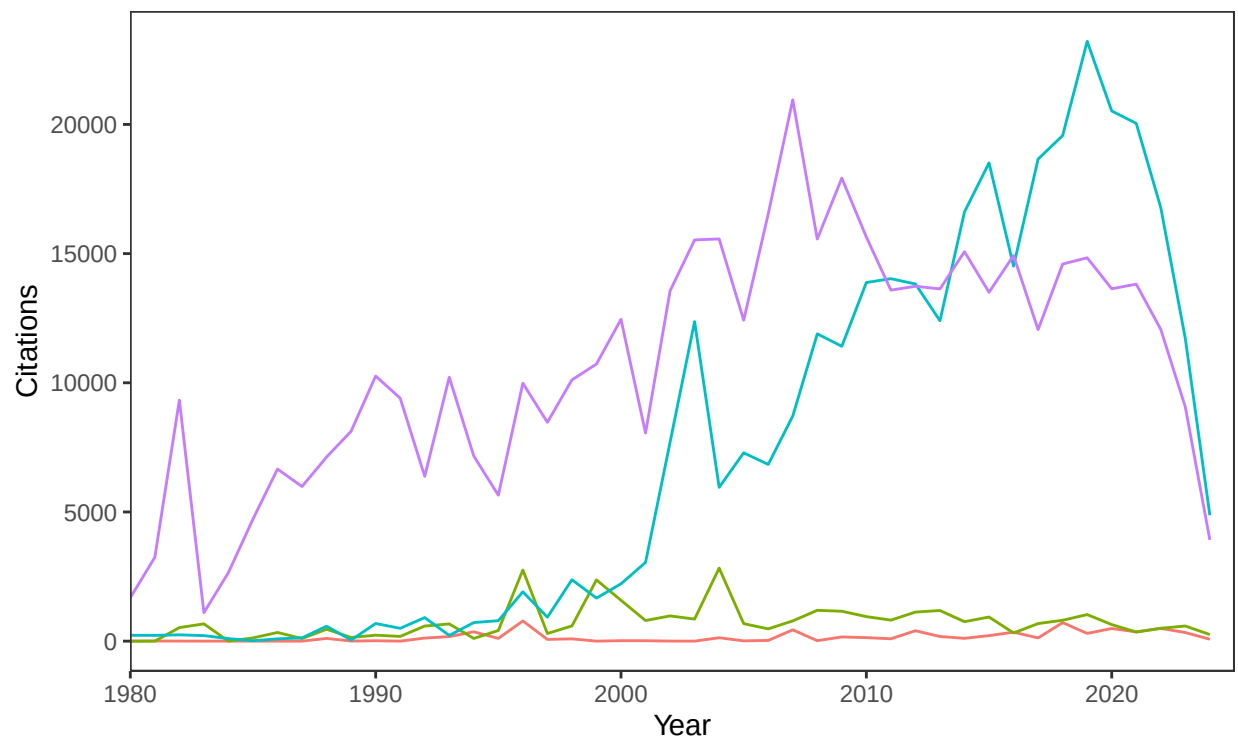
3.1 Top Domains



3.2 Domains over Time



Domain — Health Sciences — Life Sciences — Physical Sciences — Social Sciences



Domain — Health Sciences — Life Sciences — Physical Sciences — Social Sciences

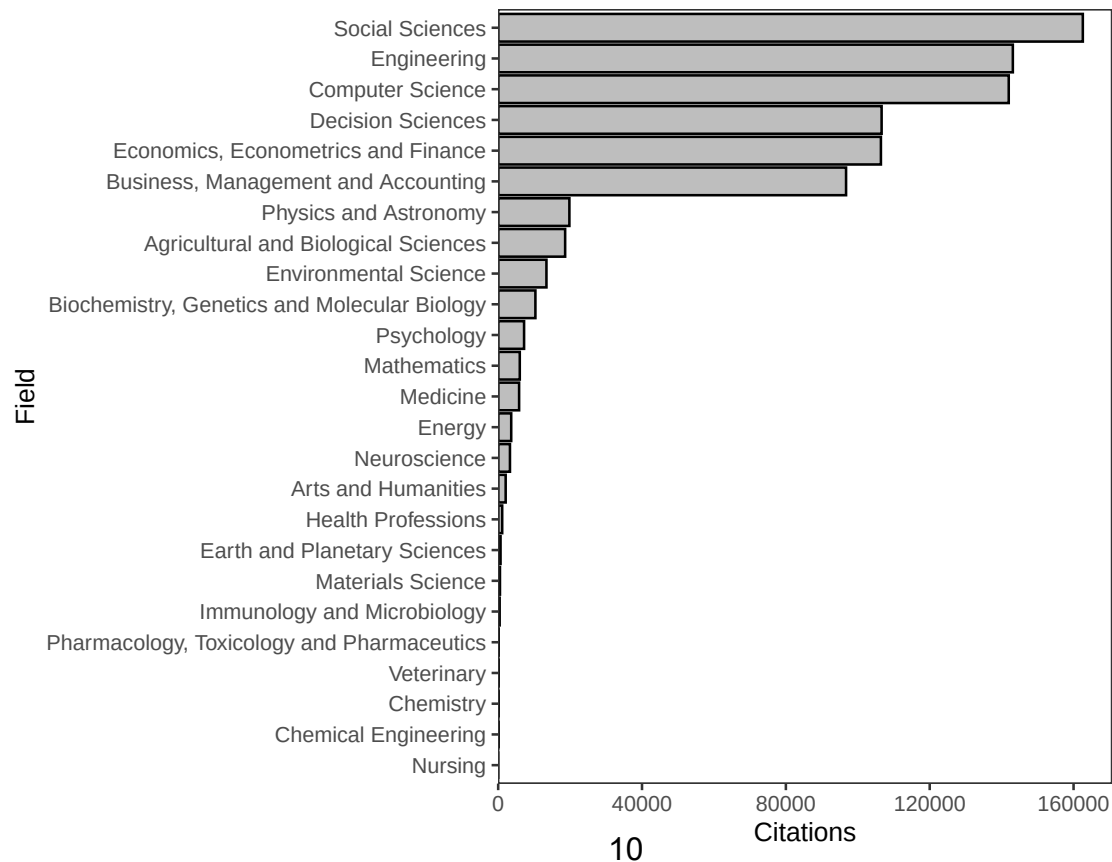
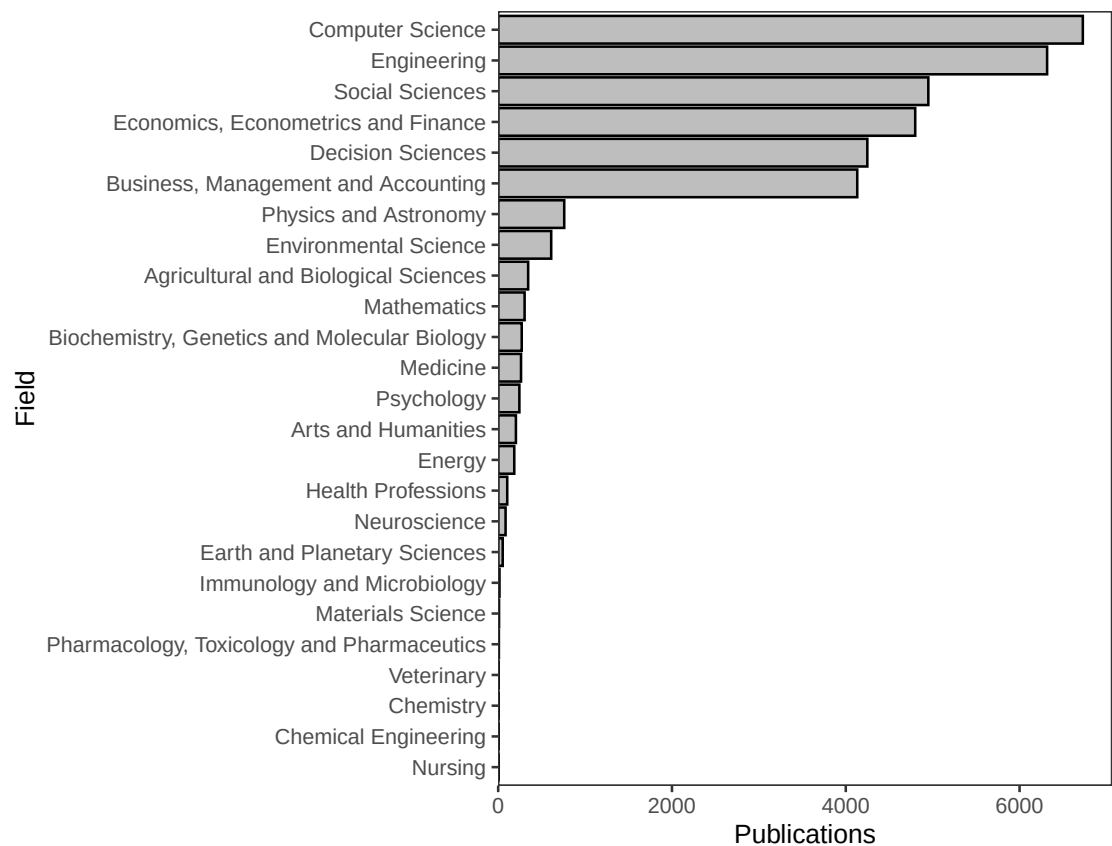
4. Field Overview

4.1 List of Fields

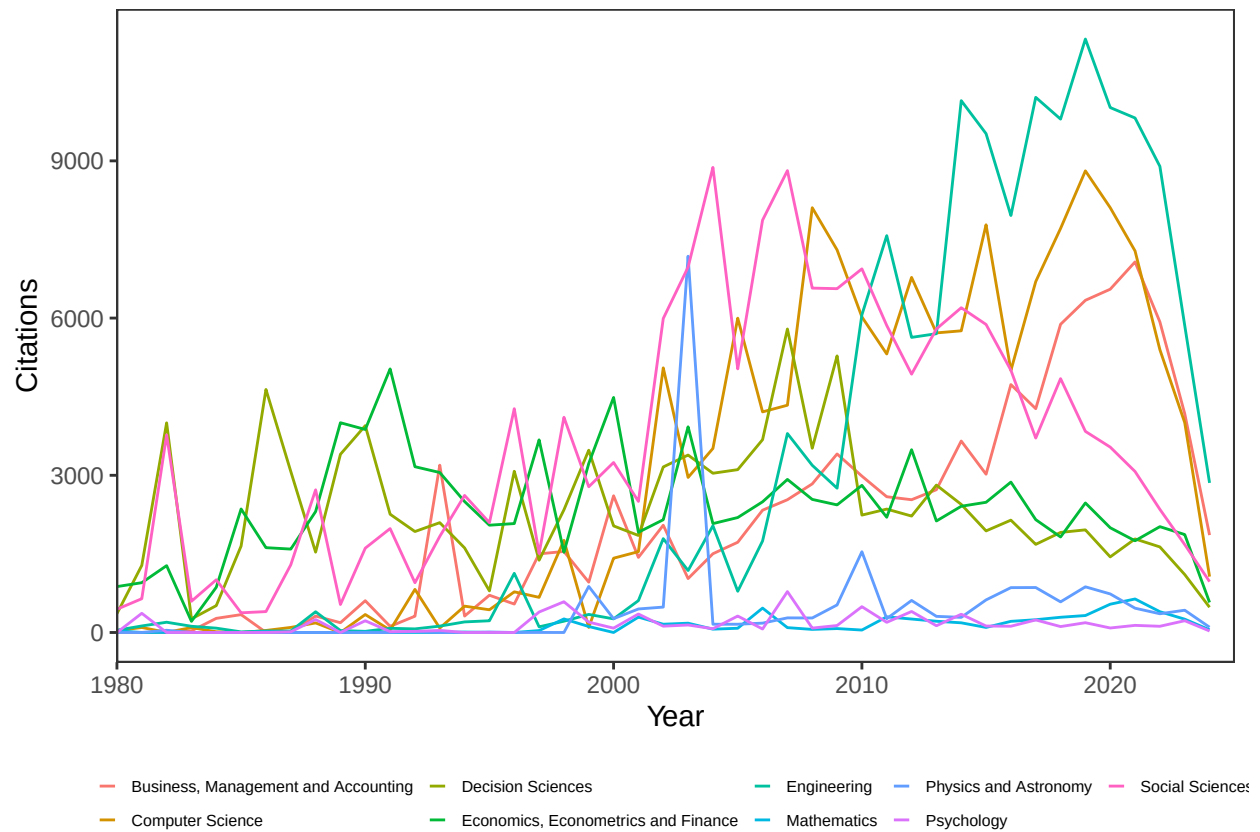
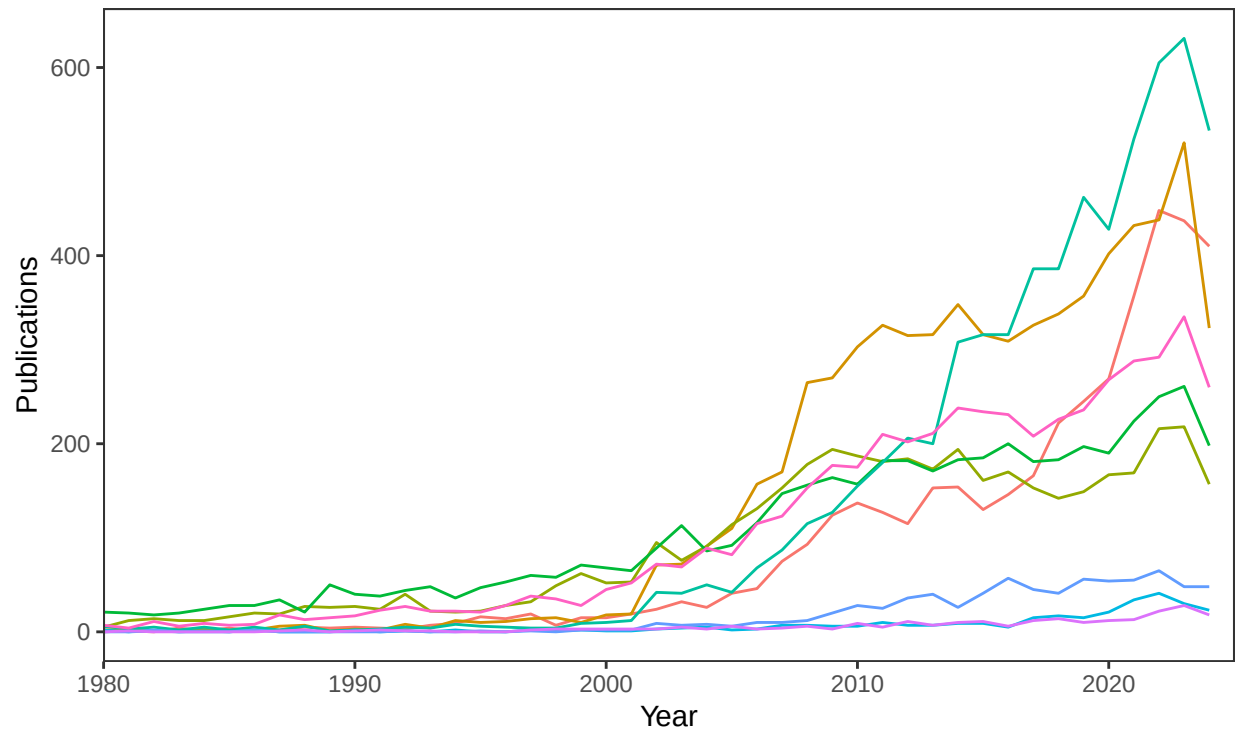
Table 4: List of Fields by OpenAlex.

Field
Agricultural and Biological Sciences
Arts and Humanities
Biochemistry, Genetics and Molecular Biology
Business, Management and Accounting
Chemical Engineering
Chemistry
Computer Science
Decision Sciences
Earth and Planetary Sciences
Economics, Econometrics and Finance
Energy
Engineering
Environmental Science
Health Professions
Immunology and Microbiology
Materials Science
Mathematics
Medicine
Neuroscience
Nursing
Pharmacology, Toxicology and Pharmaceutics
Physics and Astronomy
Psychology
Social Sciences
Veterinary

4.2 Top Fields



4.3 Fields over Time



4.4 Distribution of Fields across Queries

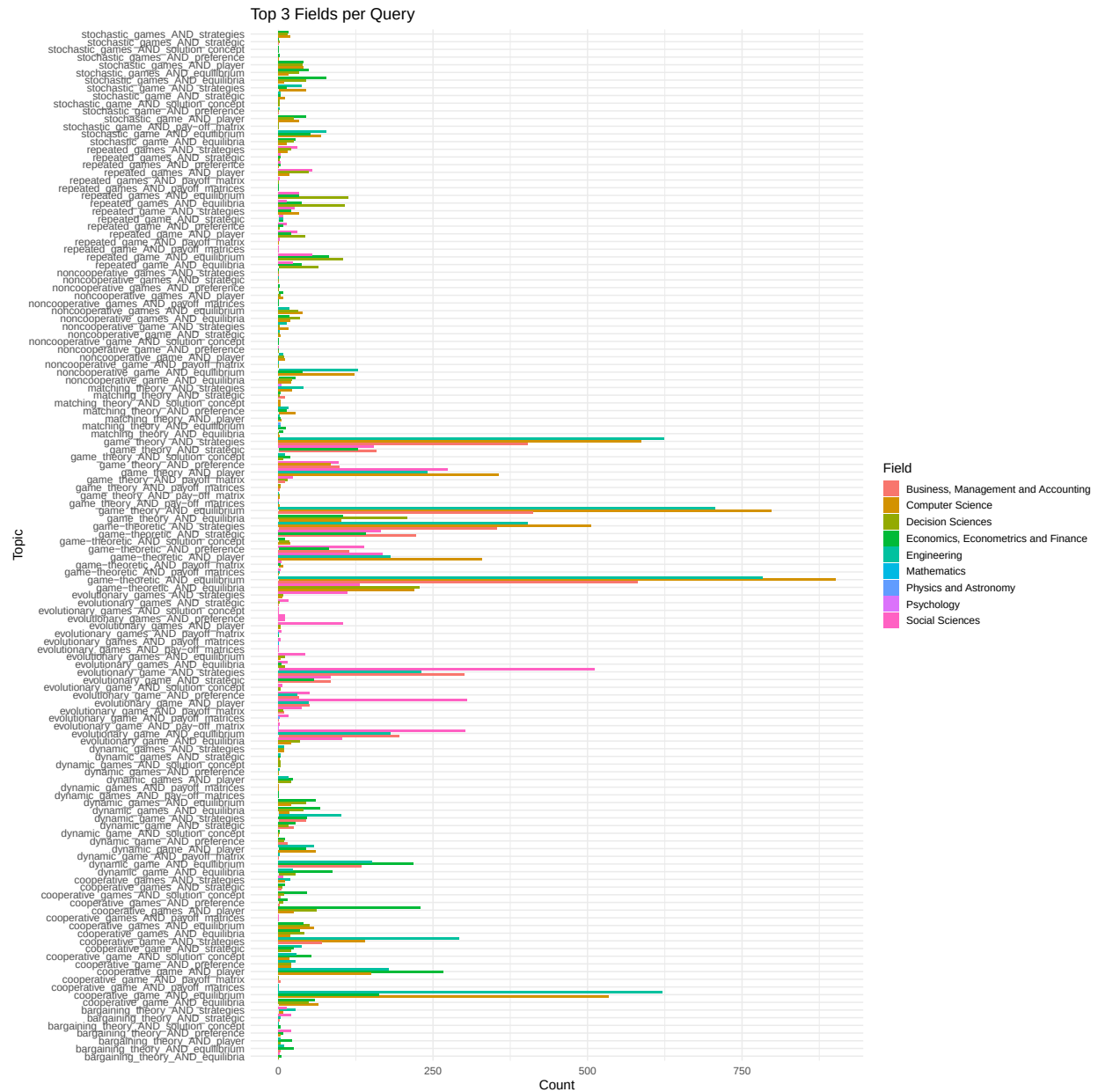
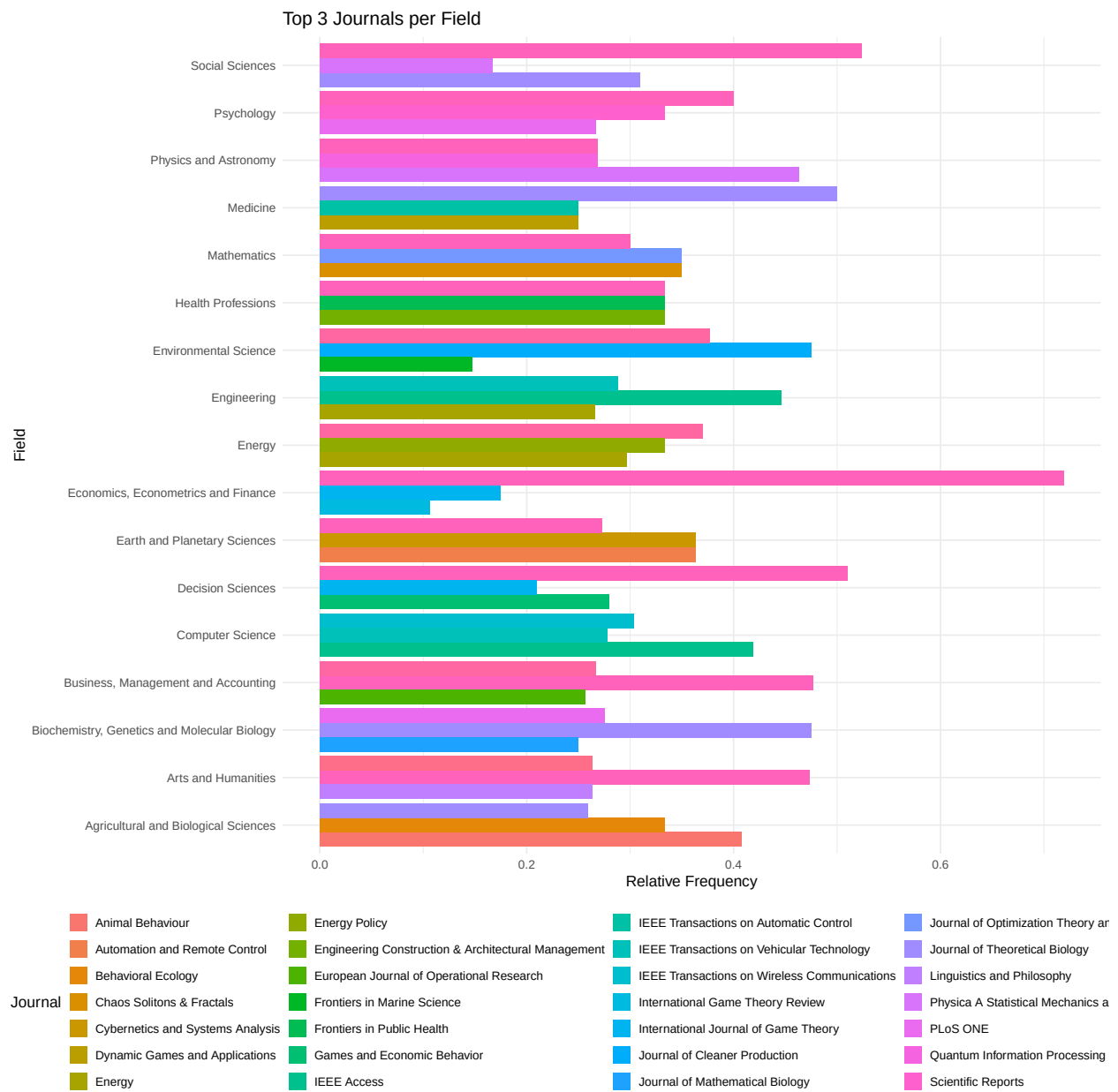


Table 5: Top Journals per Field

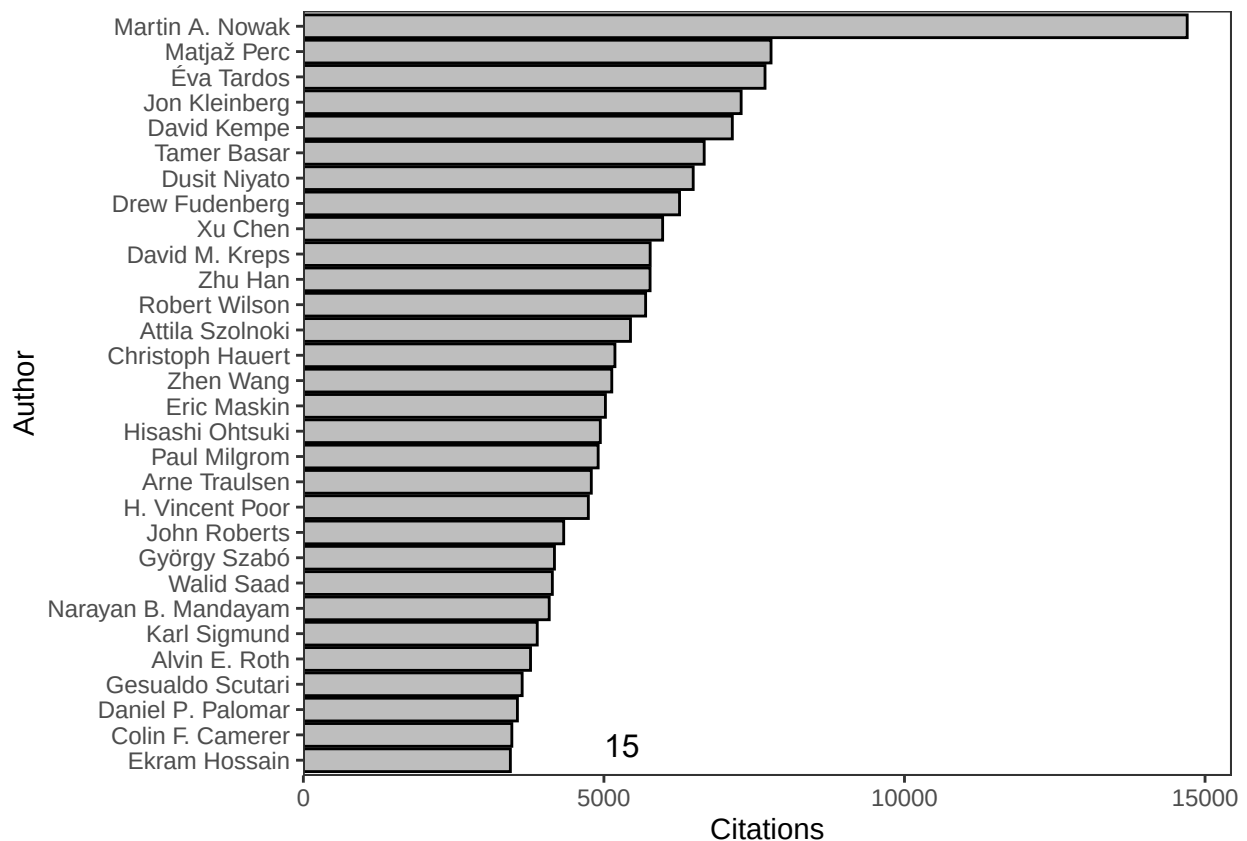
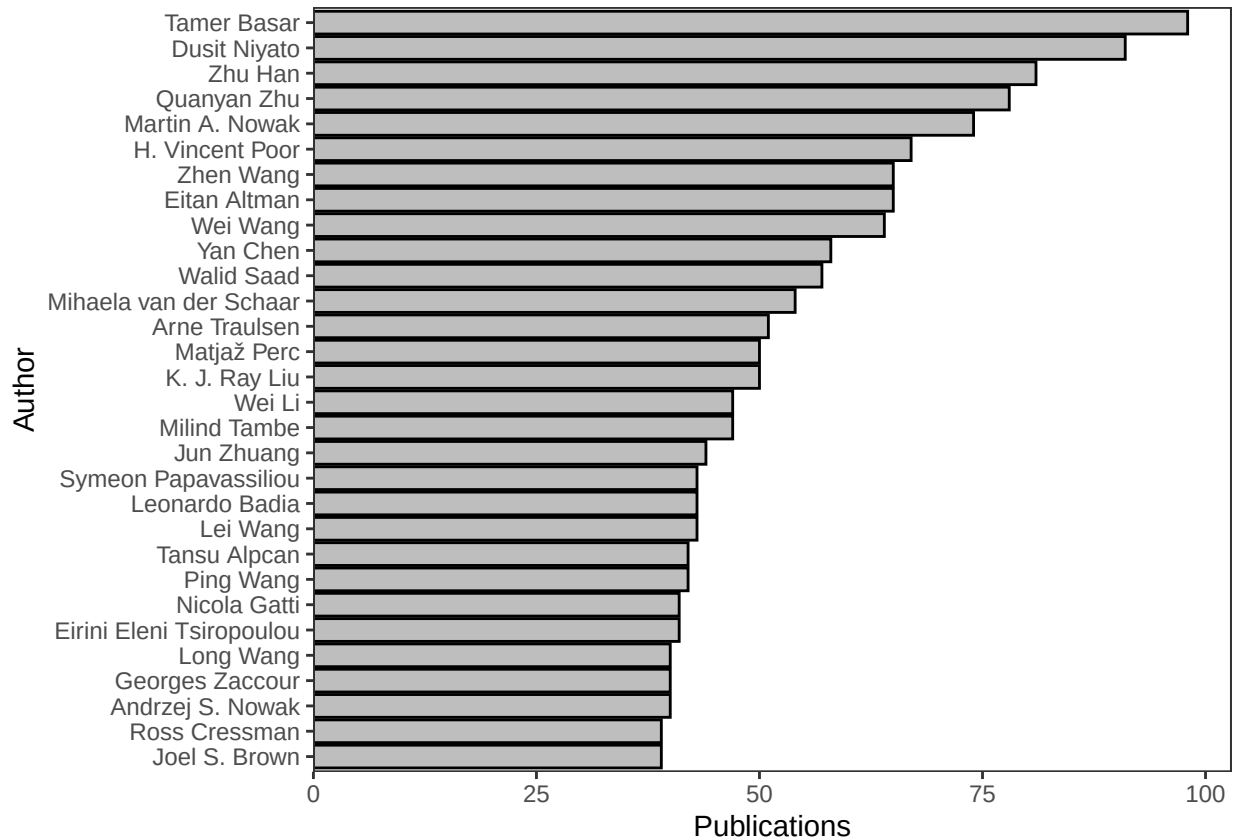
Field	Top Journals per Field (Descending Order)		
	Journal 1	Journal 2	Journal 3
Agricultural and Biological Sciences	Animal Behaviour (22)	Behavioral Ecology (18)	Journal of Theoretical Biology (14)
Arts and Humanities	SSRN Electronic Journal (9)	Linguistics and Philosophy (5)	Synthese (5)
Biochemistry, Genetics and Molecular Biology	Journal of Theoretical Biology (19)	PLoS ONE (11)	Journal of Mathematical Biology (10)
Business, Management and Accounting	SSRN Electronic Journal (225)	Sustainability (126)	European Journal of Operational Research (121)
Computer Science	IEEE Access (113)	IEEE Transactions on Wireless Communications (82)	IEEE Transactions on Vehicular Technology (75)
Decision Sciences	SSRN Electronic Journal (226)	Games and Economic Behavior (124)	International Journal of Game Theory (93)
Earth and Planetary Sciences	Automation and Remote Control (4)	Cybernetics and Systems Analysis (4)	SSRN Electronic Journal (3)
Economics, Econometrics and Finance	SSRN Electronic Journal (445)	International Journal of Game Theory (108)	International Game Theory Review (66)
Energy Engineering	Sustainability (10) IEEE Access (141)	Energy Policy (9) IEEE Transactions on Vehicular Technology (91)	Energy (8) Energy (84)
Environmental Science	Journal of Cleaner Production (29)	Sustainability (23)	Frontiers in Marine Science (9)
Health Professions	Engineering Construction & Architectural Management (4)	Frontiers in Public Health (4)	SSRN Electronic Journal (4)
Mathematics	Chaos Solitons & Fractals (7)	Journal of Optimization Theory and Applications (7)	SSRN Electronic Journal (6)
Medicine	Journal of Theoretical Biology (10)	Dynamic Games and Applications (5)	IEEE Transactions on Automatic Control (5)
Physics and Astronomy	Physica A Statistical Mechanics and its Applications (31)	Quantum Information Processing (18)	SSRN Electronic Journal (18)
Psychology	SSRN Electronic Journal (6)	Scientific Reports (5)	PLoS ONE (4)
Social Sciences	SSRN Electronic Journal (257)	Journal of Theoretical Biology (152)	Physica A Statistical Mechanics and its Applications (82)

4.5 Distribution of Journals across fields



5. Authors Overview

5.1 Top Authors



5.2 Top Authors per Domain

Table 6: Top Authors in Terms of Citations per Domain

Health Sciences	Life Sciences	Physical Sciences	Social Sciences
Gerry Mackie (765)	Mart R. Gross (1681)	Jon Kleinberg (7136)	Martin A. Nowak (13532)
Alison P. Galvani (519)	Joel S. Brown (1455)	David Kempe (7132)	Matjaž Perc (7535)
Gretchen B. Chapman (519)	Hanna Kokko (1265)	Éva Tardos (7088)	Drew Fudenberg (6258)
Timothy C. Reluga (354)	Rufus A. Johnstone (1262)	Dusit Niyato (6381)	David M. Kreps (5769)
Don M. Hugie (260)	Karl Sigmund (1194)	Tamer Başar (5852)	Robert Wilson (5694)

Table 7: Top Authors in Terms of Publications per Domain

Health Sciences	Life Sciences	Physical Sciences	Social Sciences
Jan Rychtář (6)	Joel S. Brown (18)	Dusit Niyato (88)	Martin A. Nowak (70)
K. M. Ariful Kabir (5)	Ross Cressman (11)	Zhu Han (73)	Arne Traulsen (47)
Paolo Russu (5)	J. M. Cushing (9)	Tamer Başar (72)	Matjaž Perc (45)
Angelo Antoci (4)	Jan Rychtář (9)	Quanyan Zhu (66)	Zhen Wang (45)
Chris T. Bauch (4)	Gordon G. McNickle (7)	H. Vincent Poor (60)	Andrzej S. Nowak (39)

5.3 Top Authors per Field

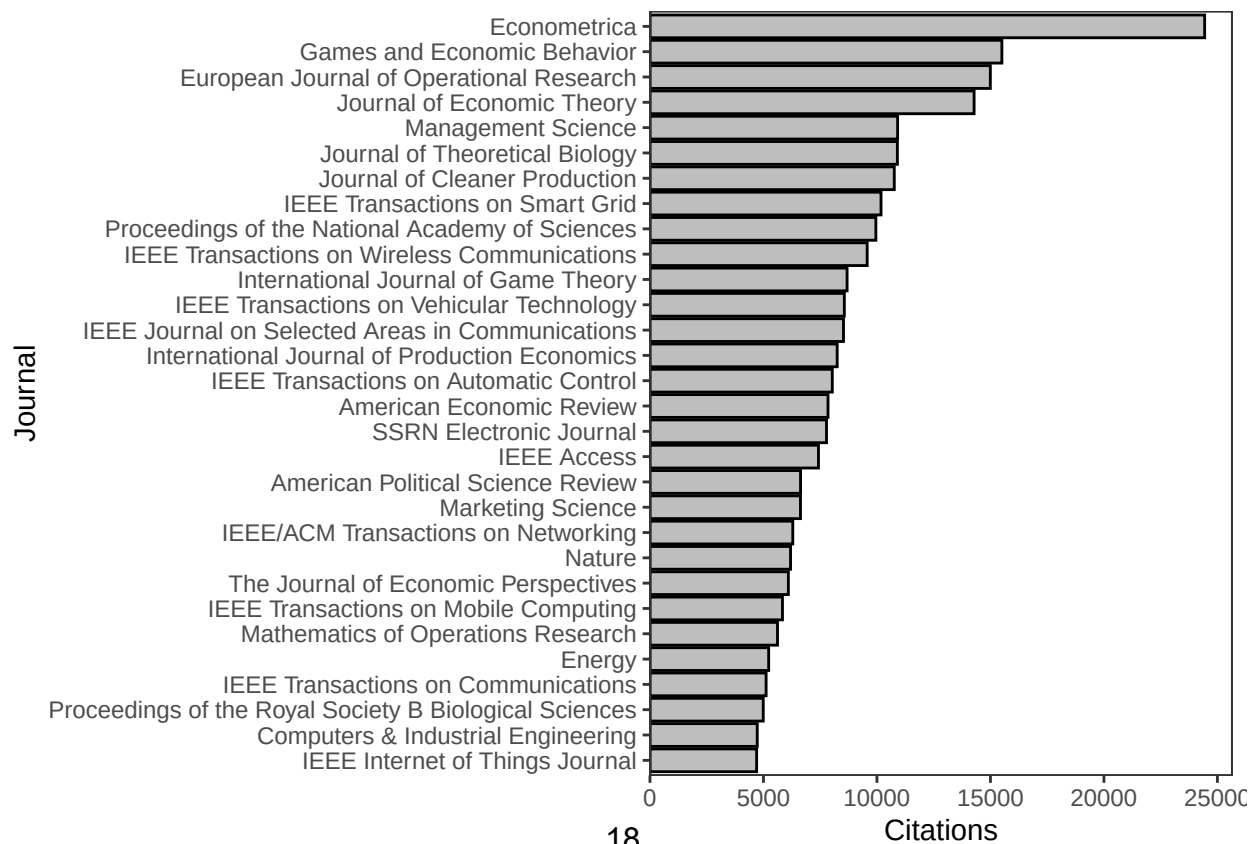
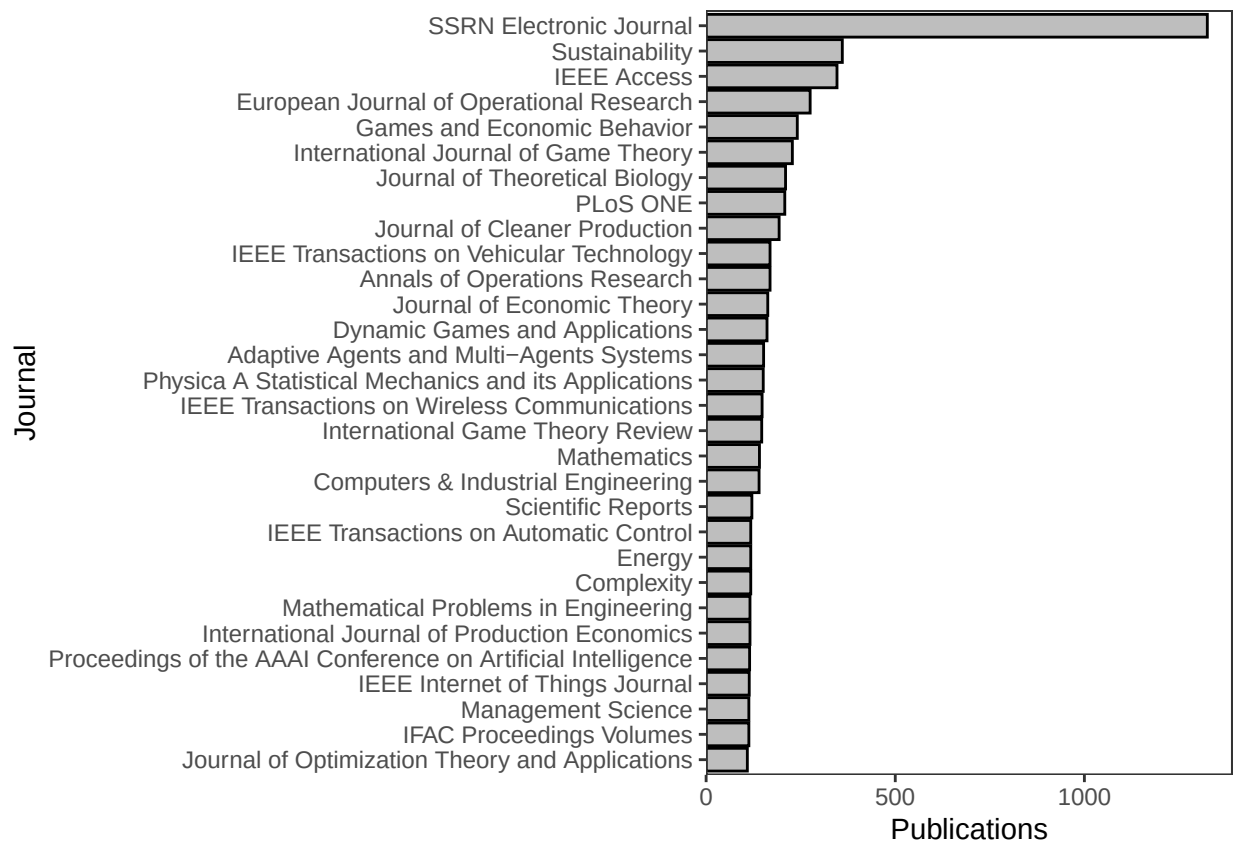
Table 8: Top Authors in Terms of Citations per Field

Field	Authors		
Business, Management and Accounting	Arvind Parkhe (3117)	Rajiv Lal (1604)	Debabrata Ghosh (1411)
Computer Science	Dusit Niyato (4837)	Xu Chen (4786)	Narayan B. Mandayam (3411)
Decision Sciences	Robert Wilson (3160)	Eric Maskin (3152)	David M. Kreps (3101)
Economics, Econometrics and Finance	Charles F. Manski (2052)	Dilip Abreu (2047)	David Pearce (2005)
Engineering	Vincent W. S. Wong (3052)	Tamer Başar (2956)	Robert Schober (2932)
Mathematics	Kenneth J. Pienta (508)	David Axelrod (451)	Robert Axelrod (451)
Physics and Astronomy	Jon Kleinberg (7111)	David Kempe (7088)	Éva Tardos (7088)
Psychology	Herbert Gintis (411)	Geoffrey F. Miller (399)	Peter M. Todd (368)
Social Sciences	Martin A. Nowak (13206)	Matjaž Perc (7529)	Attila Szolnoki (5442)

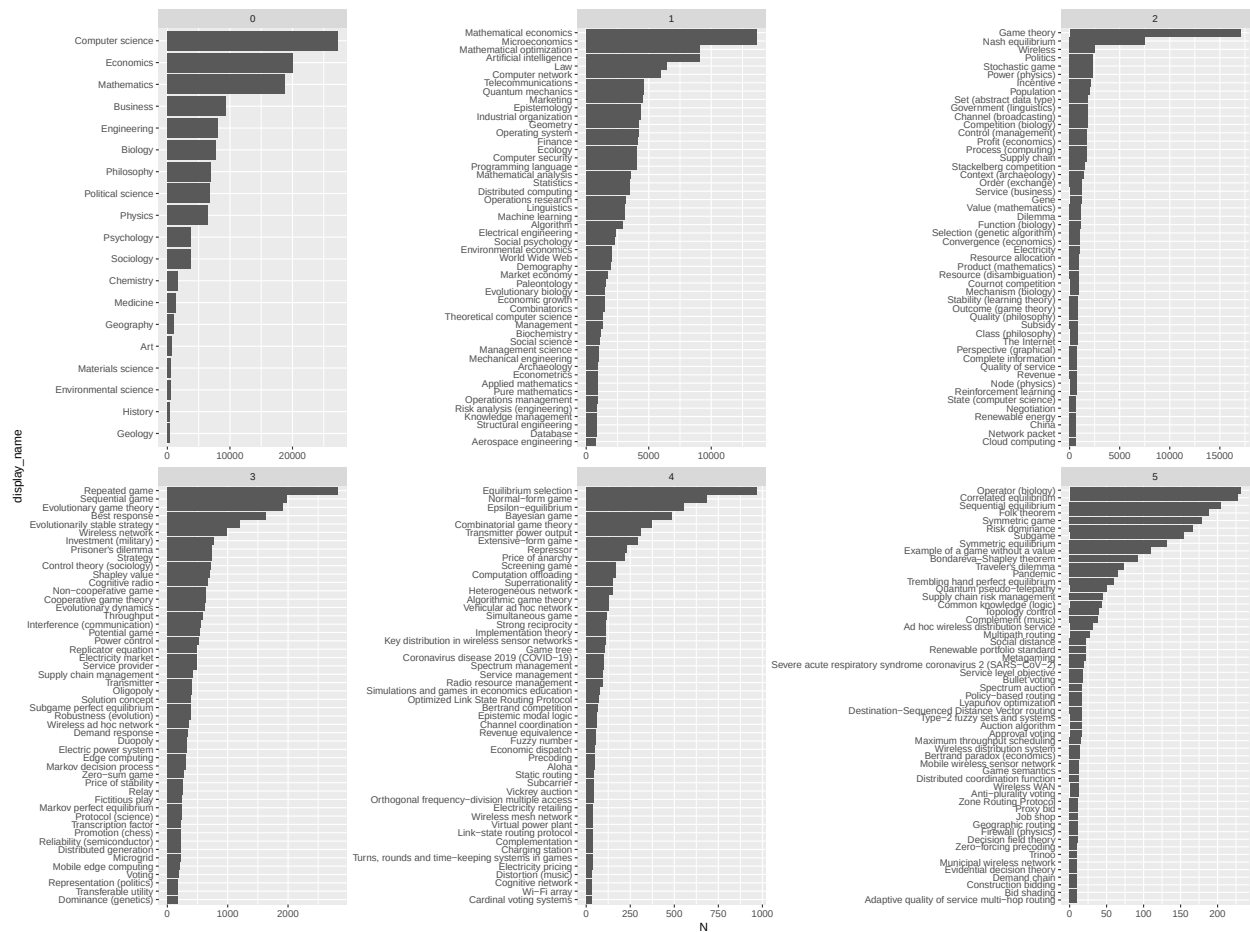
Table 9: Top Authors in Terms of Publications per Field

Field	Authors		
Business, Management and Accounting	Salma Karray (19)	Ata Allah Taleizadeh (18)	Ruiliang Yan (18)
Computer Science	Dusit Niyato (49)	Tamer Başar (36)	Charles Kamhoua (32)
Decision Sciences	Eilon Solan (21)	Jason R. Marden (21)	János Flesch (21)
Economics, Econometrics and Finance	Andrzej S. Nowak (38)	René van den Brink (35)	S.H. Tijs (30)
Engineering	Walid Saad (42)	Zhu Han (41)	Dusit Niyato (38)
Mathematics	K. M. Ariful Kabir (7)	Andrzej Świerniak (5)	Michał Krześlak (5)
Physics and Astronomy	Yan Chen (15)	K. J. Ray Liu (9)	Azhar Iqbal (8)
Psychology	Kevin McGee (4)	Teong Leong Chuah (3)	Timothy Merritt (3)
Social Sciences	Martin A. Nowak (67)	Arne Traulsen (47)	Matjaž Perc (44)

6. Journal Overview



7. Distribution of Concepts



8. Keywords

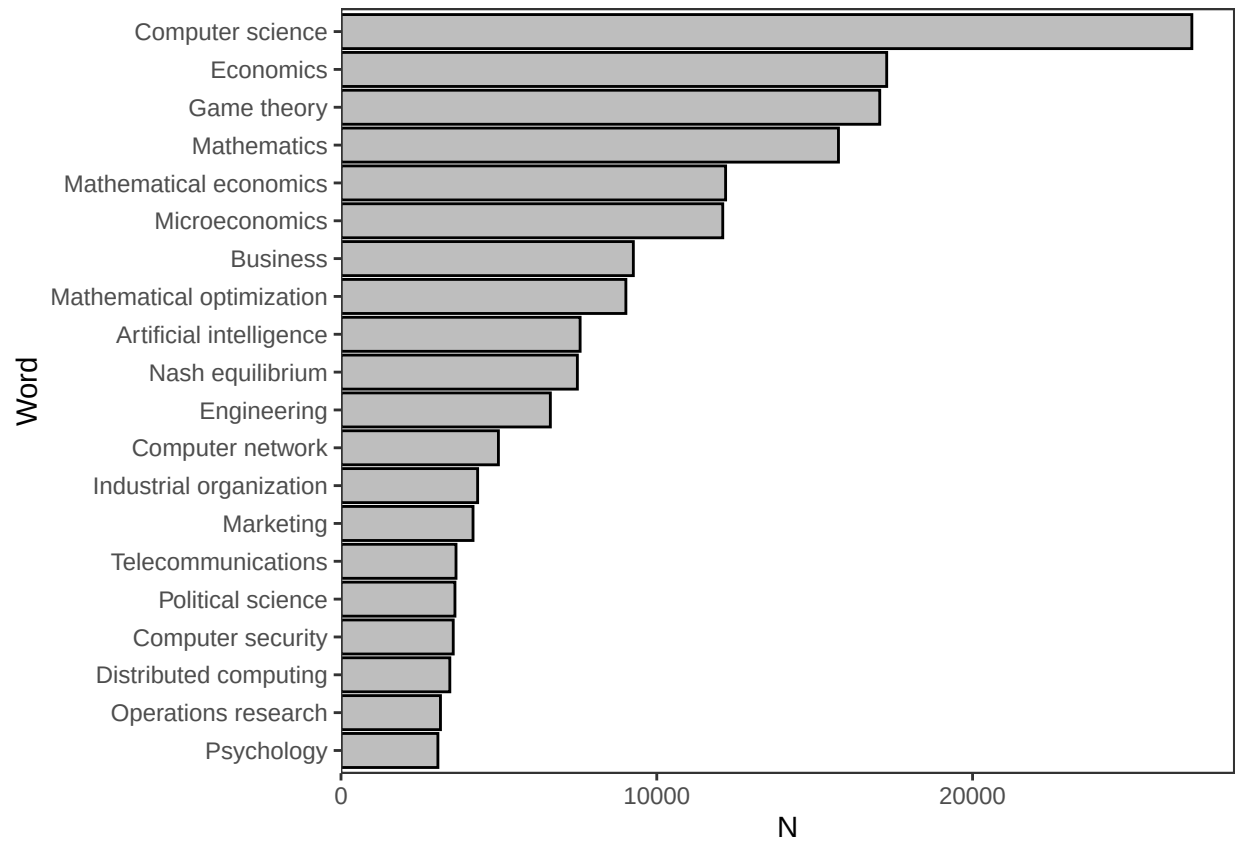


Table 10: Top Keywords

keywords	N
Computer science	26944
Economics	17281
Game theory	17061
Mathematics	15753
Mathematical economics	12180
Microeconomics	12089
Business	9258
Mathematical optimization	9021
Artificial intelligence	7572
Nash equilibrium	7485
Engineering	6630
Computer network	4985
Industrial organization	4327
Marketing	4181
Telecommunications	3642
Political science	3608
Computer security	3554
Distributed computing	3446
Operations research	3150
Psychology	3069